ISRO 2025: Topic-Wise Practice Questions

TOPIC 5: DATABASE MANAGEMENT SYSTEMS
Based on 2023 ISRO Questions: 30, 31, 32, 33, 34

Coverage Analysis: This topic had 5 questions (6.25%) in 2023 paper, focusing on SQL operations, ACID properties, normalization, and complex query scenarios.

Key Concepts Tested: SQL query execution, database transactions, normalization forms, relational algebra operations, complex database scenarios.

SQL QUERIES & OPERATIONS (Questions 1-15)

1. Consider table T(X,Y) with initial record (1,1). After inserting records 32 times with X=MX+1, Y=2*MY+1, what is SELECT Y FROM T WHERE X=4:
(a) 7
(b) 15
(c) 31
(d) 63

2. SQL statement "SELECT * FROM R, S" is equivalent to:
(a) SELECT * FROM R NATURAL JOIN S
(b) SELECT * FROM R CROSS JOIN S
(c) SELECT * FROM R INNER JOIN S
(d) SELECT * FROM R OUTER JOIN S

3. Which SQL query deletes teachers from departments located in building 'IT':
(a) DELETE FROM teacher WHERE dept_name IN 'IT'
(b) DELETE FROM department WHERE building='IT'
(c) DELETE FROM teacher WHERE dept_name IN (SELECT dept_name FROM department WHERE building='IT')
(d) DELETE FROM teacher WHERE building='IT'

4. SELECT DISTINCT removes:
(a) Duplicate rows
(b) NULL values
(c) Empty strings
(d) All of above

5. GROUP BY clause is used with:
(a) Aggregate functions
(b) ORDER BY only
(c) WHERE clause only
(d) UPDATE statement

6. HAVING clause filters:

(a) Individual rows before grouping
(b) Groups after GROUP BY
(c) All rows in table
(d) Columns in SELECT

7. Which is correct SQL join syntax:
(a) SELECT * FROM A JOIN B ON A.id = B.id
(b) SELECT * FROM A, B WHERE A.id = B.id
(c) SELECT * FROM A INNER JOIN B ON A.id = B.id
(d) All of above

8. LEFT OUTER JOIN returns:
(a) All rows from left table
(b) All rows from right table
(c) Common rows only
(d) All rows from both tables

9. Correlated subquery:
(a) References outer query columns
(b) Executes independently
(c) Cannot use EXISTS
(d) Faster than joins

10. EXISTS operator returns:
(a) TRUE if subquery returns any row
(b) FALSE if subquery empty
(c) NULL if subquery has NULL
(d) Both (a) and (b)

11. UNION operator:
(a) Combines rows from two tables
(b) Removes duplicates by default
(c) Requires same number of columns
(d) All of above

12. Difference between UNION and UNION ALL:
(a) UNION removes duplicates, UNION ALL doesn't
(b) UNION ALL is faster
(c) UNION sorts results
(d) Both (a) and (b)

13. Nested query execution:
(a) Inner query executes first
(b) Outer query executes first
(c) Simultaneous execution
(d) Depends on optimizer

14. SQL aggregate functions include:

(a) COUNT, SUM, AVG
(b) MAX, MIN
(c) GROUP_CONCAT, STRING_AGG
(d) All of above

15. Window functions differ from aggregate functions because they:
(a) Don't require GROUP BY
(b) Provide row-level results
(c) Can use ORDER BY within function
(d) All of above

DATABASE DESIGN & NORMALIZATION (Questions 16-25)

16. First Normal Form (1NF) requires:
(a) Atomic values in each cell
(b) No repeating groups
(c) Primary key defined
(d) All of above

17. Second Normal Form (2NF) eliminates:
(a) Partial functional dependencies
(b) Transitive dependencies
(c) Multi-valued dependencies
(d) Join dependencies

18. Third Normal Form (3NF) eliminates:
(a) Partial dependencies
(b) Transitive dependencies
(c) Multi-valued dependencies
(d) Join dependencies

19. Boyce-Codd Normal Form (BCNF) is:
(a) Stronger version of 3NF
(b) Same as 3NF
(c) Weaker than 3NF
(d) Independent of 3NF

20. Fourth Normal Form (4NF) eliminates:
(a) Functional dependencies
(b) Multi-valued dependencies
(c) Join dependencies
(d) Partial dependencies

21. Fifth Normal Form (5NF) is concerned with:
(a) Join dependencies
(b) Multi-valued dependencies
(c) Functional dependencies
(d) Partial dependencies

22. Denormalization is done to:
(a) Improve query performance
(b) Reduce storage space
(c) Simplify updates
(d) Eliminate redundancy

23. Functional dependency A → B means:
(a) A determines B uniquely
(b) B determines A uniquely
(c) A and B are independent
(d) A and B are same

24. Armstrong's axioms include:
(a) Reflexivity, Augmentation, Transitivity
(b) Union, Decomposition, Pseudo-transitivity
(c) Both (a) and (b)
(d) Neither (a) nor (b)

25. Closure of attribute set:
(a) All attributes functionally determined by the set
(b) Minimal set of attributes
(c) Primary key attributes
(d) Foreign key attributes

TRANSACTION MANAGEMENT & CONCURRENCY (Questions 26-40)

26. ACID properties stand for:
(a) Atomicity, Consistency, Isolation, Durability
(b) Atomicity, Concurrency, Isolation, Dependency
(c) Association, Consistency, Isolation, Durability
(d) Atomicity, Consistency, Independence, Durability

27. Atomicity property ensures:
(a) Transaction executes completely or not at all
(b) Database remains consistent
(c) Concurrent transactions don't interfere
(d) Changes persist after commit

28. Consistency property ensures:
(a) All transactions execute atomically
(b) Database integrity constraints maintained
(c) Concurrent access controlled
(d) Committed changes survive failures

29. Isolation property ensures:
(a) Transaction effects invisible until commit
(b) Database remains consistent

(c) Changes survive system failures
(d) Transactions execute atomically

30. Durability property requires changes persist:
(a) Until next transaction
(b) During system operation only
(c) Even after system failures
(d) Only in memory

31. Serializability ensures:
(a) Transactions execute serially
(b) Concurrent execution equivalent to serial execution
(c) No deadlocks occur
(d) Faster execution

32. Two-phase locking protocol:
(a) Growing phase followed by shrinking phase
(b) Prevents deadlocks completely
(c) Allows lock upgrades anytime
(d) Uses timestamp ordering

33. Deadlock occurs when:
(a) Transactions wait for each other cyclically
(b) System runs out of memory
(c) Too many concurrent transactions
(d) Database becomes inconsistent

34. Deadlock prevention methods:
(a) Wait-die, wound-wait
(b) Banker's algorithm
(c) Timeout-based abortion
(d) Both (a) and (c)

35. Timestamp-based concurrency control:
(a) Assigns timestamps to transactions
(b) Orders operations by timestamps
(c) Prevents deadlocks
(d) All of above

36. Optimistic concurrency control:
(a) Assumes conflicts are rare
(b) Validates at commit time
(c) Aborts conflicting transactions
(d) All of above

37. Isolation levels include:
(a) Read Uncommitted, Read Committed
(b) Repeatable Read, Serializable

(c) Both (a) and (b)
(d) None of above

38. Dirty read occurs when:
(a) Transaction reads uncommitted changes
(b) Transaction reads committed changes
(c) Transaction reads its own changes
(d) Transaction reads NULL values

39. Phantom read occurs when:
(a) New rows appear between reads
(b) Updated rows change between reads
(c) Deleted rows disappear between reads
(d) All of above

40. Write-ahead logging (WAL):
(a) Log records written before data changes
(b) Ensures durability
(c) Supports transaction rollback
(d) All of above

DATABASE ARCHITECTURE & STORAGE (Questions 41-50)

41. Three-schema architecture includes:
(a) External, Conceptual, Internal
(b) Logical, Physical, View
(c) User, System, Storage
(d) Application, Database, File

42. Data independence means:
(a) Changes in schema don't affect applications
(b) Data stored independently
(c) Applications independent of DBMS
(d) Database independent of OS

43. B+ tree index provides:
(a) Sequential access to data
(b) Random access to data
(c) Both sequential and random access
(d) Hash-based access

44. Hash index is best for:
(a) Range queries
(b) Equality queries
(c) Sorted access
(d) Sequential scan

45. Clustered index:

(a) Determines physical order of data
(b) Can have multiple per table
(c) Stores pointers to data
(d) Faster for all queries

46. Database buffer management:
(a) Keeps frequently used pages in memory
(b) Uses LRU replacement policy
(c) Reduces disk I/O
(d) All of above

47. RAID levels provide:
(a) Redundancy for fault tolerance
(b) Improved performance
(c) Cost-effective storage
(d) All of above

48. Database recovery uses:
(a) Transaction logs
(b) Database backups
(c) Checkpoints
(d) All of above

49. Query optimization involves:
(a) Choosing efficient execution plans
(b) Cost-based decisions
(c) Index utilization
(d) All of above

50. Distributed database characteristics:
(a) Data distributed across multiple sites
(b) Appears as single database to users
(c) Requires distributed transaction management
(d) All of above

ADVANCED DATABASE CONCEPTS (Questions 51-55)

51. NoSQL databases are suitable for:
(a) Unstructured data
(b) High scalability requirements
(c) Flexible schema needs
(d) All of above

52. CAP theorem states that distributed systems can guarantee at most:
(a) 2 out of 3: Consistency, Availability, Partition tolerance
(b) All 3: Consistency, Availability, Partition tolerance
(c) 1 out of 3: Consistency, Availability, Partition tolerance
(d) None of the above

53. Data warehouse characteristics:
(a) Subject-oriented, integrated, time-variant
(b) Non-volatile, decision support
(c) Both (a) and (b)
(d) Neither (a) nor (b)

54. OLTP vs OLAP:
(a) OLTP for transactions, OLAP for analysis
(b) OLTP normalized, OLAP denormalized
(c) OLTP current data, OLAP historical data
(d) All of above

55. Data mining techniques include:
(a) Classification, clustering, association rules
(b) Decision trees, neural networks
(c) Regression, anomaly detection
(d) All of above

ANSWER KEY - DATABASE MANAGEMENT SYSTEMS

1-10: (a), (b), (c), (a), (a), (b), (d), (a), (a), (d)
11-20: (d), (d), (a), (d), (d), (d), (a), (b), (a), (b)
21-30: (a), (a), (a), (c), (a), (a), (a), (b), (a), (c)
31-40: (b), (a), (a), (d), (d), (d), (c), (a), (a), (d)
41-50: (a), (a), (c), (b), (a), (d), (d), (d), (d), (d)
51-55: (d), (a), (c), (d), (d)

Critical Study Focus for September 14th:

1. **SQL Operations (Q1-15)**: Master complex queries, joins, subqueries, and set operations
2. **Normalization (Q16-25)**: Understand all normal forms and when to apply them
3. **Transactions (Q26-40)**: Focus on ACID properties, concurrency control, and isolation levels
4. **Database Design (Q41-50)**: Know indexing, storage structures, and optimization

Important SQL Patterns to Practice:
- Nested subqueries with EXISTS/IN operators
- Complex JOIN operations (INNER, OUTER, CROSS)
- Aggregate functions with GROUP BY and HAVING
- Correlated subqueries and window functions

Key Concepts for Quick Review:

- ACID properties and their practical implications
- Normal forms progression (1NF → 2NF → 3NF → BCNF → 4NF → 5NF)
- Concurrency control mechanisms and deadlock handling
- Index types and their use cases

Study Strategy: Focus heavily on SQL query writing and execution as ISRO tests practical database operations more than theoretical concepts.